CENG 463 – Introduction to Machine Learning
Homework 1: Water Resource Risk Classification
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1. Introduction

Project Objective

The goal of this project is to classify the *Water Resource Risk Category* of countries (0–4 scale) using hydrological and environmental indicators from the **WRI – Aqueduct Water Risk Atlas** dataset. We performed feature engineering, trained five ML models, evaluated them, tuned hyperparameters using GridSearchCV, and analyzed feature importance.

Dataset Description

The dataset contains the following variables:

- **bws_score** – Baseline Water Stress
- **gtd_score** – Groundwater Depletion
- **drr_score** – Drought Risk
- **rfr_score** – River Flood Risk
- **sev_score** – Seasonal Variability
- **w_awr_def_tot_cat** – *Target class* (Water Risk Category, 0–4)

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2. Feature Engineering

2.1 Composite Water Stress Index (CWSI)

CWSI shows an extremely strong correlation with Baseline Water Stress (0.96) and a high correlation with the target risk category (0.87). This indicates that combining baseline stress, groundwater depletion, and drought risk produces a more informative variable than using them individually. In other words, CWSI successfully summarizes overall water pressure and becomes one of the most predictive engineered features.

$$\text{CWSI} = 0.5 \cdot bws_score + 0.3 \cdot gtd_score + 0.2 \cdot drr_score$$

Purpose:

- Measures combined water pressure from stress, groundwater depletion, and drought.
- Higher score = higher overall water stress.

2.2 Seasonal–Flood Interaction (SFI)

SFI displays a moderate positive correlation (0.25) with the target variable and strong correlations with rfr_score (flood risk) and sev_score (seasonal variability). This confirms that SFI captures regions where hydrological instability and flooding occur together.

$$\text{SFI} = sev_score \times rfr_score$$

Purpose:

- Captures regions with both high seasonal variability *and* flood risk.
- High values indicate combined climatic instability.

Discussion

CWSI clearly emerges as the strongest predictor among all features, correlating 0.87 with the water risk category. This confirms that combining drought risk, groundwater depletion, and baseline water stress better explains high-risk regions than individual indicators. SFI contributes complementary hydrological information by identifying areas with both high seasonal variability and flood risk. The engineered features enrich the dataset without causing multicollinearity problems, and they help model training by providing more meaningful composite indicators.

	bws_score	gtd_score	drr_score	rfr_score	sev_score	CWSI	SFI	w_awr_def_tot_cat
0	1.628	0.390	0.517	1.620	1.944	1.035	3.150	0.000
1	0.973	1.692	3.663	4.001	1.295	1.727	5.183	2.000
2	0.000	1.301	2.237	0.152	0.625	0.838	0.095	0.000
3	5.000	1.226	1.603	0.000	0.855	3.188	0.000	3.000
4	0.000	0.793	0.239	0.000	2.916	0.286	0.000	0.000

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3. Model Training and Evaluation

Five baseline models were trained as required:

- Random Forest
- Support Vector Machine (SVM)
- KNN
- Gaussian Naive Bayes
- Logistic Regression

3.1 Baseline Accuracy Table

	Model	Accuracy
0	Random Forest	0.915
1	KNN	0.804
2	SVM	0.765
3	Logistic Regression	0.673
4	Gaussian NB	0.605

The baseline evaluation shows that the Random Forest classifier achieved the highest accuracy (0.915), significantly outperforming the other models. KNN (0.804) and SVM (0.765) followed, benefiting from feature scaling but still lagging behind tree-based methods. Logistic Regression (0.673) performed moderately, reflecting limitations in modeling nonlinear interactions. Gaussian Naive Bayes yielded the lowest accuracy (0.605), likely due to its independence assumptions being violated because hydrological indicators are strongly correlated.

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4. Hyperparameter Optimization

4.1 Methodology

Hyperparameter tuning was performed using GridSearchCV with 5-fold cross-validation and accuracy as the scoring metric. For each classifier, an appropriate hyperparameter grid was defined, and the best estimator was evaluated on the test set.

4.2 Tuned Hyperparameters Table

	Model	Baseline Accuracy	Tuned Accuracy	Best Params
0	Random Forest	0.915	0.912	{'clf__max_depth': 20, 'clf__min_samples_split...
1	KNN	0.804	0.875	{'clf__n_neighbors': 3, 'clf__weights': 'dista...
2	SVM	0.765	0.802	{'clf__C': 10, 'clf__gamma': 'scale'}
3	Logistic Regression	0.673	0.671	{'clf__C': 10, 'clf__solver': 'lbfgs'}

	Model	Baseline Accuracy	Tuned Accuracy	Best Params
4	Gaussian NB	0.605	0.605	{'clf__var_smoothing': 1e-09}

4.3 Mandatory Classification Report

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============================================================
BEST MODEL: Random Forest
============================================================
Test Accuracy: 0.912

Classification Report:
              precision    recall  f1-score   support

         0.0       0.96      0.96      0.96       147
         1.0       0.88      0.92      0.90       174
         2.0       0.88      0.86      0.87       174
         3.0       0.91      0.90      0.90       193
         4.0       0.94      0.93      0.93       223

    accuracy                           0.91       911
   macro avg       0.91      0.91      0.91       911
weighted avg       0.91      0.91      0.91       911
```

4.4 Discussion

Hyperparameter tuning produced varying impacts across models. KNN showed the largest improvement, increasing from 0.804 to 0.875 when using distance-based weighting and fewer neighbors. SVM also improved slightly, reaching 0.802 with C=10. Random Forest did not improve; the tuned model (0.912) performed slightly worse than the baseline (0.915), likely due to over-regularization when limiting max_depth. Logistic Regression and Gaussian NB showed

no meaningful improvement, consistent with their simpler model capacity. Overall, tuning primarily benefited instance-based and margin-based models (KNN and SVM).

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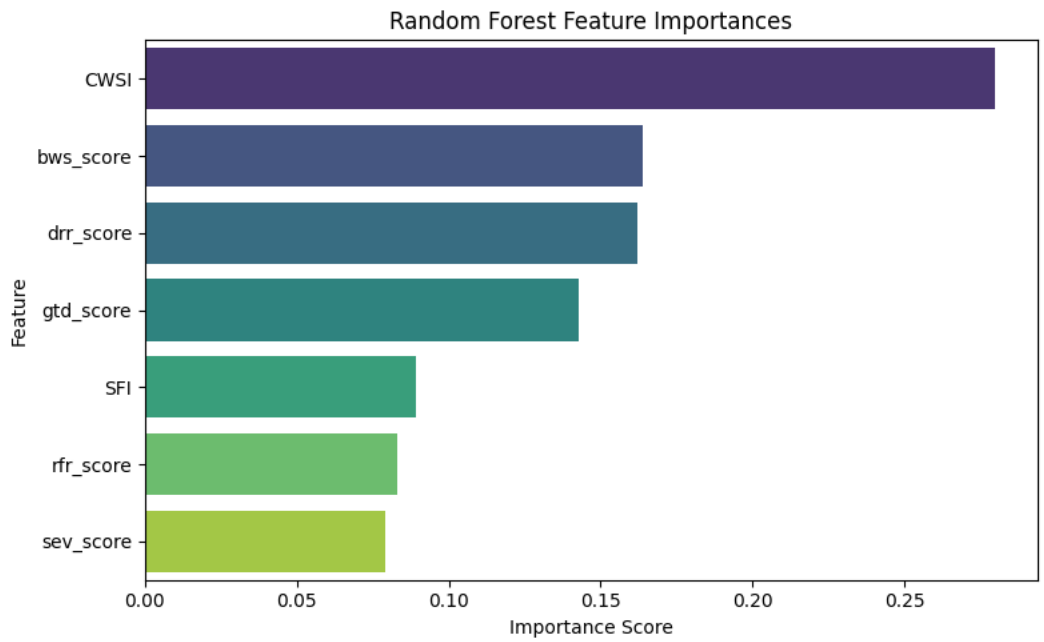
5. Feature Importance Analysis

5.1 Feature Importance Results

The Random Forest model ranks CWSI as the most important feature (0.280), followed by bws_score (0.164), drr_score (0.162), and gtd_score (0.143). SFI (0.089), rfr_score, and sev_score contribute less but still provide supplementary information.



5.3 Interpretation

CWSI being the most important feature confirms that chronic water stress factors dominate risk classification. Baseline water stress and drought risk also play major roles, consistent with their strong correlations with the target. gtd_score contributes moderately, highlighting the relevance of groundwater depletion. SFI, rfr_score, and sev_score have lower importance individually because much of their effect is already captured within composite variables such as CWSI and SFI. Overall, feature importance analysis verifies that long-term scarcity indicators are more predictive of water risk than short-term variability factors.

6. Final Conclusion

The models evaluated in this study show that tree-based and instance-based algorithms outperform linear and probabilistic approaches for water risk classification. Although **Random Forest achieved the highest baseline accuracy (0.915)**, hyperparameter tuning produced the **best tuned performance with KNN (0.875)**.

Feature engineering significantly strengthened the models, with CWSI emerging as the dominant predictor of water risk, followed by baseline stress and drought risk. These results highlight that chronic scarcity factors are the primary drivers of water resource vulnerability. Future work could explore boosting algorithms such as XGBoost, incorporate additional hydrological indicators, or utilize spatial-temporal modeling to further improve predictive performance.